

PHISHING INCIDENT ANALYSIS & FORENSIC INVESTIGATION

INCIDENT OVERVIEW

- **What happened:** A targeted phishing attack compromised employee credentials.
- **When:** Incident occurred on May 2021.
- **Who was affected:** Employees in the finance department and IT administrators.
- **Impact:** Unauthorized access to internal systems and data leakage.



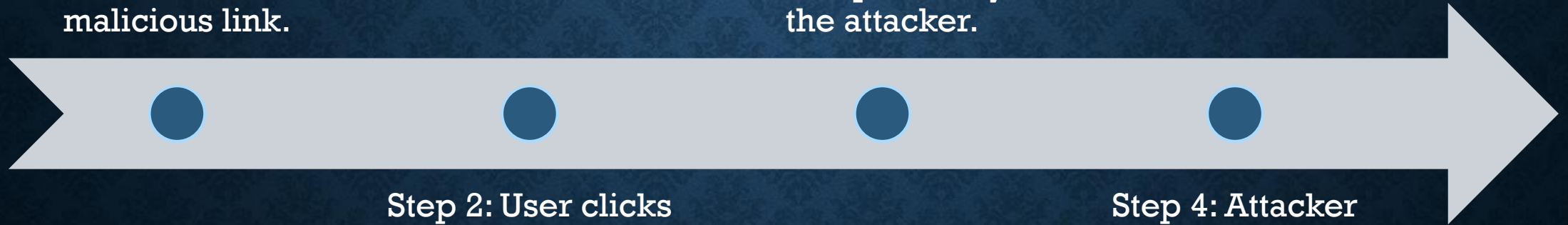
ATTACK METHODOLOGY – PHISHING WORKFLOW

Step 1: Attacker
sends deceptive
email with
malicious link.

Step 3: User inputs
credentials, which
are captured by
the attacker.

Step 2: User clicks
the link, landing
on a fake login
page.

Step 4: Attacker
uses stolen
credentials to
access internal
network.



FORENSIC ANALYSIS – TOOL 1: VOLATILITY

Purpose:

- Memory forensics to identify suspicious activity.

Process:

- Load memory dump into Volatility.
- Search for suspicious processes and injected code.
- Analyze browser artifacts for phishing URLs.

Findings:

- Presence of unusual processes and phishing domain URLs in memory.

```
root@NeilFoxDe1XPS: ~/Volatility3
Volatility 3 Framework 2.0.0 PDB scanning finished
```

PID	PPID	ImageFileName	Offset(V)	Threads	Handles	SessionId	Wow64	CreateTime	ExitTime	File output
4	0	System	0xbe0a574ac040 153	-	N/A	False	2021-05-20 07:28:25.000000	N/A	Disabled	
312	4	smss.exe	0xbe0a58d18800	2	-	N/A	False	2021-05-20 07:28:25.000000	N/A	Disabled
428	412	csrss.exe	0xbe0a59215800	11	-	0	False	2021-05-20 07:28:36.000000	N/A	Disabled
492	312	smss.exe	0xbe0a598cd080	0	-	1	False	2021-05-20 07:28:39.000000	2021-05-20 07:28:40.000000	Disabled
500	412	wininit.exe	0xbe0a598c2080	1	-	0	False	2021-05-20 07:28:39.000000	N/A	Disabled
516	492	csrss.exe	0xbe0a59900280	15	-	1	False	2021-05-20 07:28:39.000000	N/A	Disabled
584	492	winlogon.exe	0xbe0a5994e800	6	-	1	False	2021-05-20 07:28:40.000000	N/A	Disabled
636	500	services.exe	0xbe0a599a2800	9	-	0	False	2021-05-20 07:28:41.000000	N/A	Disabled
652	500	lsass.exe	0xbe0a599cb080	8	-	0	False	2021-05-20 07:28:41.000000	N/A	Disabled
740	636	svchost.exe	0xbe0a59891800	25	-	0	False	2021-05-20 07:28:42.000000	N/A	Disabled
816	636	svchost.exe	0xbe0a59a60800	17	-	0	False	2021-05-20 07:28:43.000000	N/A	Disabled
924	584	dwm.exe	0xbe0a59ab84c0 11	-	1	False	2021-05-20 07:28:44.000000	N/A	Disabled	
956	636	svchost.exe	0xbe0a59a54800	93	-	0	False	2021-05-20 07:28:44.000000	N/A	Disabled
1020	636	svchost.exe	0xbe0a59a50800	29	-	0	False	2021-05-20 07:28:44.000000	N/A	Disabled
80	636	svchost.exe	0xbe0a59a4e800	38	-	0	False	2021-05-20 07:28:44.000000	N/A	Disabled
496	636	svchost.exe	0xbe0a59a4c800	23	-	0	False	2021-05-20 07:28:44.000000	N/A	Disabled
800	636	svchost.exe	0xbe0a59a4a800	28	-	0	False	2021-05-20 07:28:44.000000	N/A	Disabled
1144	636	svchost.exe	0xbe0a59bbd800	28	-	0	False	2021-05-20 07:28:44.000000	N/A	Disabled
1452	636	svchost.exe	0xbe0a58de7300	8	-	0	False	2021-05-20 07:28:44.000000	N/A	Disabled
1600	636	svchost.exe	0xbe0a5963a5c0	11	-	0	False	2021-05-20 07:28:45.000000	N/A	Disabled
1684	636	spoolsv.exe	0xbe0a596bf800	14	-	0	False	2021-05-20 07:28:45.000000	N/A	Disabled
1948	636	svchost.exe	0xbe0a59cea380	15	-	0	False	2021-05-20 07:28:46.000000	N/A	Disabled

Volatility 3 Framework 2.0.0 PDB scanning finished

PID	PPID	ImageFileName	Offset(V)	Threads
4	0	System	0xbe0a574ac040 153	-
* 312	4	smss.exe	0xbe0a58d18800	2
** 492	312	smss.exe	0xbe0a598cd080	0
*** 584	492	winlogon.exe	0xbe0a5994e800	6
**** 664	584	userinit.exe	0xbe0a5a5d6800	
***** 1900	664	explorer.exe	0xbe0a5a5ed800	
***** 5384	1900	MSASCuiL.exe	0xbe0a5ae3a800	
***** 5408	1900	chrome.exe	0xbe0a5c67c080	
***** 6112	5408	chrome.exe	0xbe0a604ae200	
***** 1028	5408	chrome.exe	0xbe0a585fc800	
***** 1832	5408	chrome.exe	0xbe0a5cba3080	
***** 4112	5408	chrome.exe	0xbe0a59b65080	
***** 8116	5408	chrome.exe	0xbe0a57eb4800	
***** 4856	5408	chrome.exe	0xbe0a5cf63080	
***** 6780	5408	chrome.exe	0xbe0a5fb90300	
***** 5428	1900	vmtoolsd.exe	0xbe0a5ae07800	
***** 6648	1900	ProcessHacker.	0xbe0a5a9f4800	
**** 924	584	dwm.exe	0xbe0a59ab84c0 11	

FORENSIC DEMO SCREENSHOTS – VOLATILITY

- **Memory Dump Overview:** Screenshot of Volatility memory image loaded.
- **Suspicious Processes:** Highlight suspicious process IDs and names.
- **Browser Artifacts:** Screenshot showing phishing URLs extracted from browser memory.

FORENSIC ANALYSIS – TOOL 2: WIRESHARK

Purpose:

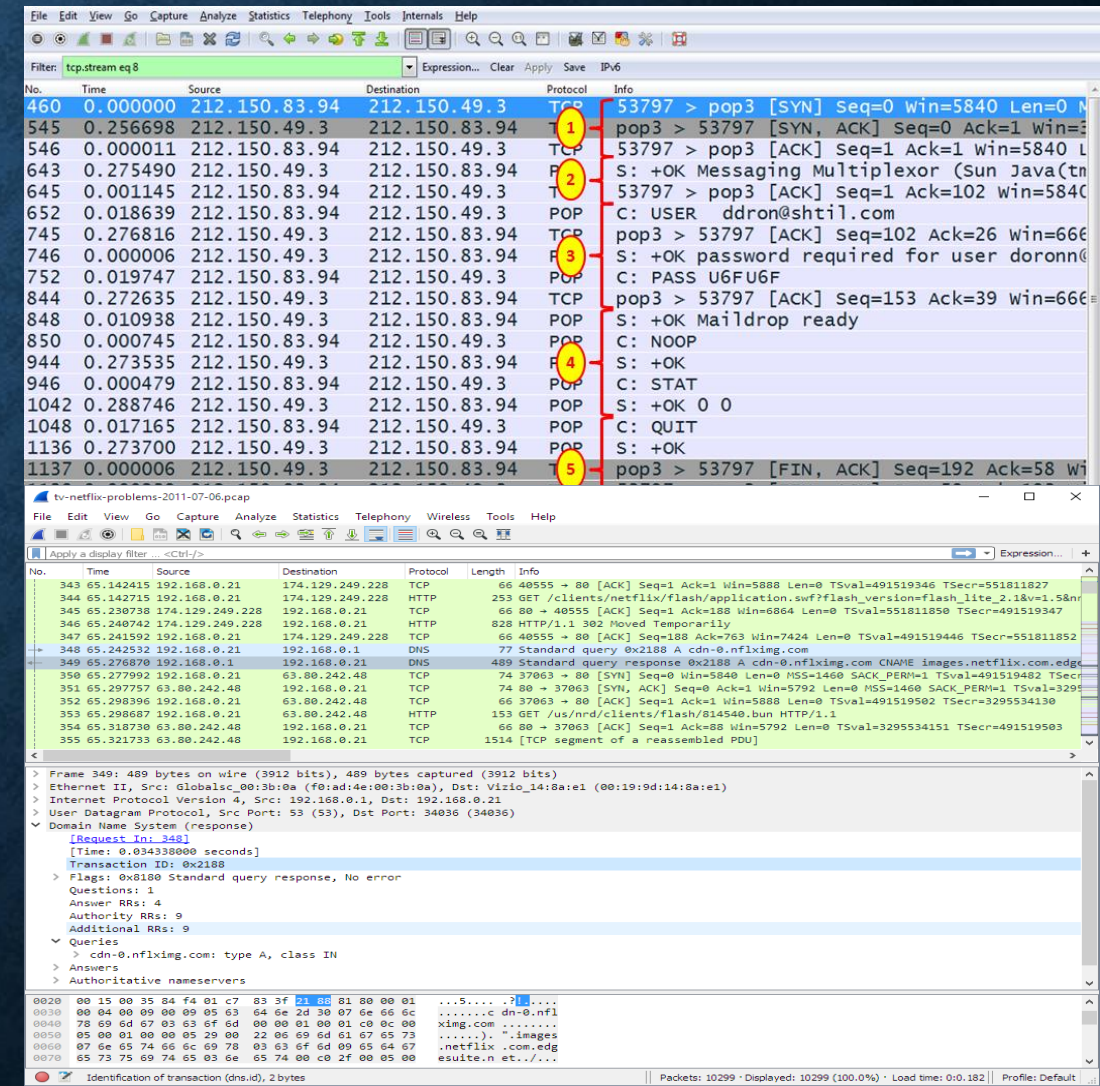
- Network traffic analysis to track phishing domain communication.

Process:

- Load packet capture from affected workstation.
- Filter DNS requests for phishing domains.
- Inspect packet payloads for malicious indicators.

Findings:

- DNS requests to phishing domains and unusual outbound traffic patterns.



LEGAL & ETHICAL CONSIDERATIONS

- **Privacy Act Compliance:** Handling of personal data during investigation.
- **Notifiable Data Breaches (NDB):** Mandatory reporting obligations.
- **User Responsibility:** Importance of employee vigilance and ethical use of IT resources.

RECOMMENDATIONS

User Awareness Training:

- Regular phishing simulations and education.

Zero Trust Security Model:

- Verify every access request, no implicit trust.

Technical Controls:

- Email filtering and URL scanning.
- Multi-factor authentication (MFA).
- Endpoint detection and response (EDR).

CONCLUSION & KEY LEARNINGS

- Early detection critical to minimizing damage.
- Memory and network forensics complement each other.
- Continuous user education and robust technical controls prevent phishing success.
- Legal compliance ensures responsible incident handling.

Q&A

- Questions?
- Thank you for your attention!